

Phase space mixing in the presence of self-gravity: spherically symmetric Vlasov-Poisson evolutions with fixed and dispersed angular momentum

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[[Erik] Provisional; to be rewritten when sections VI and VII are final.] This article is devoted to the numerical study of a spherically symmetric, collisionless kinetic gas which is trapped in an external central potential and whose self-gravity is taken into account. In the absence of self-gravity, such a gas relaxes in time to a stationary state through phase space mixing. We analyze how this process is modified by the self-gravity of the gas, considering both the reduced model in which all gas particles have the same magnitude of the angular momentum and the case in which the angular momenta are distributed over an interval. Our study is based on a particle-in-cell code, on a numerical solution of the linearized equations in action-angle variables, and on a spectral criterion which characterizes the existence of oscillating modes. We find that small perturbations of a self-consistent steady state are damped as long as the mass of the gas lies below a threshold, whereas above this threshold they give rise to an oscillation which is not damped, and we show that the threshold and the frequency of the oscillation are accurately predicted by the spectral criterion. Finally, we show that a dispersion in the angular momentum suppresses the oscillations which are present in the reduced model.

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I. INTRODUCTION

Collisionless kinetic gases provide the standard description for systems consisting of a large number of particles which interact with each other only through the mean field they generate. Prominent examples are the stars in a galaxy or in a globular cluster, and the dark matter particles in a galactic halo [1]. The state of such a gas at time t is described by its one-particle distribution function (DF) $f(t, x, p)$, a nonnegative function on the one-particle phase space which determines the number of gas particles in a small phase space volume about the point (x, p) , and whose evolution is governed by the Vlasov equation.¹ In the Newtonian gravitational case, the Vlasov equation is coupled to Poisson's equation for the gravitational potential generated by the gas, which yields the Vlasov-Poisson system, see Ref. [2] for a review of its mathematical properties. Its general relativistic counterpart is the Vlasov-Einstein system [3].

It is instructive to compare the dynamics of a self-gravitating collisionless gas with the one of a pressureless fluid (dust), which is obtained in the limit in which the velocity dispersion of the gas vanishes. The gravitational collapse of a spherical dust cloud leads to the formation of a singularity in finite time at which the density diverges, both in the Newtonian theory and in general relativity [4]; in the latter case, part of this singularity may even be visible to distant observers [5, 6]. In contrast to this, the velocity dispersion of a kinetic gas prevents the particles from focusing. In fact, the Vlasov-Poisson system admits global classical solutions for general initial data [7–9], while for the spherically symmetric Vlasov-Einstein system global solutions exist for small initial data [10] and sufficiently concentrated data collapse to a black hole [11], the threshold between these two outcomes having been explored numerically in Refs. [12–15]. Therefore, for a Newtonian collisionless gas the relevant question is not whether or not a singularity forms, but rather how the gas behaves for large times and, in particular, whether or not it relaxes to a stationary state.

A fundamental mechanism leading to such a relaxation is phase space mixing [16, 17]. To explain it, consider first a gas whose self-gravity is neglected and which is trapped in the potential well of an external potential. If the one-particle Hamiltonian is integrable, each gas particle is confined to an invariant torus, and in terms of action-angle variables (Q, J) the solution of the Vlasov equation is $f(t, Q, J) = F(Q - \Omega(J)t, J)$, where F describes the initial DF and $\Omega(J)$ denotes the frequencies associated with the torus labeled by J . Since these frequencies vary from one torus to the other, the DF is stretched along the angle directions and develops structures on finer and finer scales as time increases. As a consequence, f does not converge pointwise as $t \rightarrow \infty$. However, the macroscopic observables which are obtained by integrating f against a test function do converge, and their limit is the one associated with

¹ The Vlasov equation is also often called the collisionless Boltzmann equation in the literature.

the average of the initial DF over the angle variables, which depends only on integrals of motion. A mathematically precise formulation of this statement and its proof for rotationally symmetric potentials in $d = 1, 2, 3$ space dimensions satisfying a suitable non-degeneracy condition were given by Rioseco and Sarbach [18] (see also Refs. [19, 20]). The results in Ref. [18] include the case of a spherically symmetric gas configuration in which all the gas particles have the same value of the total angular momentum, which behaves like an effective one-dimensional system, and they explain the relaxation to a virialized state which had been observed in the numerical simulations of Ref. [21]. For the analogous phenomenon in the exterior of a black hole we refer the reader to Refs. [22, 23], and for estimates on the rate at which the observables converge to Refs. [24–26].

The generalization to the self-gravitating case, which was left open in Ref. [18], is considerably more involved, since the potential now depends on the DF, such that a perturbation of the latter induces a perturbation of the potential which, in turn, acts back on the gas. In plasma physics, where the particles interact through their mean electric field, the corresponding collective effect has been known since the work of Landau [27]: for a gas of charged particles close to a spatially homogeneous equilibrium the electric field decays in time while the DF undergoes mixing. This phenomenon, known as Landau damping, has been established rigorously at the nonlinear level by Mouhot and Villani [28]. The gravitational case differs from the electromagnetic one in two important aspects. First, the interaction is attractive, such that the collective response of the gas tends to reinforce a perturbation instead of screening it. Second, a self-gravitating gas configuration of finite mass does not possess spatially homogeneous equilibria. Instead, its steady states are spatially inhomogeneous, the gas particles are trapped in the potential well, and the frequencies of their radial motion lie in an interval which, in the situations considered in this article, is bounded and bounded away from zero. For this reason, a collective oscillation whose frequency lies below this interval is not in resonance with any gas particle and hence cannot be damped by mixing.

The question of whether small perturbations of a steady state of the gravitational Vlasov-Poisson system are damped or give rise to persistent oscillations goes back to Lynden-Bell [16] and has received considerable attention in recent years. Hadžić, Rein and Straub [29] (see also Refs. [30, 31]) showed that the essential spectrum of the operator governing the linearized dynamics about a large class of spherically symmetric steady states is determined by the radial periods of the gas particles. Using a version of the Birman-Schwinger principle which was first applied to this problem by Mathur [32], they derived a criterion for the existence of an eigenvalue in the gap below this spectrum, which corresponds to an oscillating mode that is not damped. This criterion is formulated in terms of a family of bounded operators acting on functions of the radius. Related criteria for the existence of oscillating modes and estimates for their number, which also allow for an attractive external potential, have been given by Moreno, Rioseco and Van Den Bosch [33] for steady states whose DF depends only on the energy. For the reduced system in which all the gas particles have the same magnitude of the angular momentum and move around a central point mass, Hadžić, Rein, Schrecker and Straub [34] proved the following dichotomy for steady states of sufficiently small mass whose DF is proportional to $(E_0 - E)^k$ for energies E below the cutoff energy E_0 : the solutions of the linearized system are damped if $k > 1$, whereas an oscillating mode exists if $1/2 < k \leq 1$. Quantitative decay rates for the linearized system about steady states with a central point mass and a distribution of angular momenta have been obtained in Ref. [35], and for the nonlinear system with an external Kepler potential, phase mixing of spherically symmetric data over long time scales has been established in Ref. [36]. On the numerical side, Ramming and Rein [37] and Straub [38] have performed particle-in-cell simulations of the spherically symmetric Vlasov-Poisson system which show that, depending on the steady state, its perturbations either oscillate without damping or decay. Finally, we mention that weakly damped non-radial modes of spherical stellar systems have been computed in Refs. [39, 40] by means of the matrix method.

In this article, we study numerically the transition between mixing and persistent oscillations for a spherically symmetric collisionless gas which is trapped in an external central potential and whose self-gravity is taken into account. We consider two situations. The first one is the reduced model in which all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum, which has a two-dimensional phase space and is the setting considered in Refs. [18, 34]. In the second situation the angular momenta of the gas particles are distributed over an interval. For the external potential we choose the isochrone potential, for which the action-angle variables are known explicitly, and the potential generated by a point mass. Our analysis is based on three independent tools: (i) a particle-in-cell (PIC) code which solves the nonlinear Vlasov-Poisson system, (ii) a numerical solution of the equations linearized about a self-consistent steady state, formulated in the action-angle variables of the latter, and (iii) the numerical computation of the largest eigenvalue $\lambda(\omega)$ of the operator of Mathur, which measures the gain of the self-consistent loop in which a perturbation of the potential with frequency ω induces a perturbation of the density which, in turn, generates a potential. An oscillating mode with frequency below the frequency band of the gas particles exists if and only if the value λ_{edge} of this gain at the lower edge of the band is larger than one.

The main results of this article can be summarized as follows. First, as long as the mass of the gas is sufficiently small compared to the mass which generates the external potential, a small perturbation of a self-consistent steady state is damped, and the frequency and the damping rate of the collective response obtained from the PIC simulations

agree with those obtained from the linearized equations to within one percent. Second, for the reduced model there is a threshold mass above which a discrete mode emerges below the frequency band, such that the perturbation oscillates without being damped. We show that this threshold is characterized by the condition $\lambda_{edge} = 1$, and that the frequency of the mode predicted by the equation $\lambda(\omega) = 1$ agrees with the one measured in the simulations. Third, for steady states of small mass with $k \leq 1$, the oscillating mode whose existence is guaranteed by the results in Ref. [34] lies so close to the edge of the band that it cannot be distinguished from the decaying part of the response over the time span of our simulations; instead, we observe an algebraic decay proportional to $t^{-(k+1)}$ for all the values of k we have considered. Fourth, a dispersion in the angular momentum removes the divergence of $\lambda(\omega)$ at the edge of the band which is responsible for the oscillating modes with $k \leq 1$ in the reduced model, such that these modes disappear when the mass of the gas is small enough, and it suppresses the coherent response which is present when all the gas particles have the same angular momentum. **[(Erik) The last statement relies on the operator with dispersion and on the free solution; the PIC runs with dispersion of section VII are still to be done.]**

Particle methods have a long tradition in the numerical study of spherically symmetric collisionless systems, both in the Newtonian case [37, 38] and in the relativistic one, where they have been used to analyze the collapse of star clusters and the critical behavior at the threshold of black hole formation [12–14, 41, 42]; see Refs. [43–45] for introductions to the PIC method, Ref. [46] for a higher-order implementation in the context of plasma physics, and Ref. [47] for a convergence proof of a PIC scheme for the spherically symmetric Vlasov-Einstein system. Measuring the response of the gas to a small perturbation with such a method is delicate, because the signal has to be separated from the noise which is due to the finite number of computational particles. In this article we address this problem by placing the particles on a regular lattice in the action-angle variables of the self-consistent steady state, by encoding the perturbation in the weights of the particles instead of their positions (as in Refs. [48, 49]), and by subtracting a reference run without perturbation. We also show that the observables need to be computed with the action-angle variables belonging to the total potential, since those of the external potential alone leave a time-independent residue which could be mistaken for an undamped mode.

The remainder of this article is organized as follows. In section II we formulate the Vlasov-Poisson system with an external potential in spherical symmetry, for the reduced model with fixed angular momentum as well as for a distribution of angular momenta, and we introduce the action-angle variables and the observables used throughout this article. In section III we review the mixing phenomenon in the absence of self-gravity, derive the linearized equations about a steady state and introduce the gain $\lambda(\omega)$ of the self-consistent loop. Our numerical methods are described in section IV, and their validation is presented in section V. The results for the reduced model are discussed in section VI, and those for a distribution of angular momenta in section VII. Conclusions and an outlook to future work are given in section VIII. Technical details regarding the numerical computation of the action-angle variables are included in Appendix A.

II. THE VLASOV-POISSON SYSTEM IN SPHERICAL SYMMETRY

In this section we formulate the Vlasov-Poisson system with an external potential and reduce it to spherical symmetry, first for a gas whose particles have arbitrary angular momenta and then for the reduced model in which all of them have the same magnitude of the angular momentum. Next, we introduce the dimensionless variables used in our numerical computations, the action-angle variables associated with a time-independent potential and, finally, the observables on which our analysis of the mixing phenomenon is based.

We consider a collisionless gas consisting of identical particles which are subject to a time-independent external potential Φ_{ext} and to the gravitational potential Φ_{gas} generated by the gas itself. Since all the gas particles have the same mass, it is convenient to normalize the DF such that $f(t, x, p)d^3x d^3p$ is the mass of the gas contained in the volume element $d^3x d^3p$ of phase space, where from now on p denotes the momentum per unit mass.² With these conventions, the Vlasov-Poisson system consists of the Vlasov equation

$$\frac{\partial f}{\partial t} + p \cdot \nabla_x f - (\nabla_x \Phi) \cdot \nabla_p f = 0, \quad \Phi := \Phi_{ext} + \Phi_{gas}, \quad (1)$$

and the Poisson equation

$$\Delta \Phi_{gas} = 4\pi G \rho, \quad \rho(t, x) := \int_{\mathbb{R}^3} f(t, x, p) d^3p, \quad (2)$$

² This is equivalent to working in units for which the mass of each gas particle is one, as in Ref. [18].

where G denotes Newton's constant and ρ is the mass density of the gas. To obtain a unique solution of Eq. (2) we require that $\Phi_{gas}(t, x) \rightarrow 0$ for $|x| \rightarrow \infty$.

For the external potential we choose H  non's isochrone potential [1, 50, 51]

$$\Phi_{ext}(x) = -\frac{GM}{b + \sqrt{b^2 + r^2}}, \quad r := |x|, \quad (3)$$

which is generated by a spherical matter distribution of total mass M . The parameter $b \geq 0$ is a length scale which describes the transition from a harmonic potential (when $r \ll b$) to the potential of a point particle of mass M (when $r \gg b$). For $b = 0$, Eq. (3) reduces to the Kepler potential $\Phi_{ext} = -GM/r$, which is the external potential considered in Ref. [34]. The reason for this choice is that in this potential the period of the radial motion depends only on the energy of the particle [50, 51], and that the corresponding action-angle variables are known in closed form [1]. When the self-gravity of the gas is neglected, this provides exact solutions against which our numerical methods can be validated.

A. Reduction to spherical symmetry

A spherically symmetric gas configuration is described by a DF which is invariant with respect to simultaneous rotations of x and p . Such a DF depends on (x, p) only through the radius r , the radial component $p_r := x \cdot p/r$ of the momentum and the magnitude $L := |x \wedge p|$ of the angular momentum, that is,

$$f(t, x, p) = \mathcal{F}(t, r, p_r, L). \quad (4)$$

In this case, the density ρ and the potential Φ_{gas} depend only on t and r , and the Vlasov equation (1) reduces to

$$\frac{\partial \mathcal{F}}{\partial t} + p_r \frac{\partial \mathcal{F}}{\partial r} - \frac{\partial V_L}{\partial r} \frac{\partial \mathcal{F}}{\partial p_r} = 0, \quad V_L(t, r) := \Phi(t, r) + \frac{L^2}{2r^2}, \quad (5)$$

with V_L the effective potential for the radial motion. Since no derivative with respect to L appears in Eq. (5), the gas particles with a given value of L behave like a one-dimensional gas in the potential V_L . However, particles with different values of L interact with each other through the potential Φ_{gas} , which is determined by the spherically reduced Poisson equation

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi_{gas}}{\partial r} \right) = 4\pi G \rho, \quad \rho(t, r) = \frac{2\pi}{r^2} \int_0^\infty \int_{-\infty}^\infty \mathcal{F}(t, r, p_r, L) dp_r L dL, \quad (6)$$

see for instance Ref. [21]. The solution of Eq. (6) which is regular at the center and vanishes at infinity is

$$\Phi_{gas}(t, r) = -4\pi G \int_0^\infty \frac{\rho(t, s)}{\max\{r, s\}} s^2 ds, \quad (7)$$

and its gradient is $\partial \Phi_{gas}/\partial r = Gm(t, r)/r^2$, where $m(t, r) := 4\pi \int_0^r \rho(t, s) s^2 ds$ is the mass of the gas contained in the sphere of radius r at time t .

For the following, it is useful to note that when a spherically symmetric function is integrated over phase space, the volume element can be replaced with

$$d^3x d^3p = 8\pi^2 dr dp_r L dL. \quad (8)$$

In particular, the total mass and the total energy of the gas are given by

$$M_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t, r, p_r, L) dr dp_r L dL, \quad (9)$$

$$\mathcal{E}_{gas} = 8\pi^2 \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \mathcal{F}(t, r, p_r, L) \left[\frac{p_r^2}{2} + \frac{L^2}{2r^2} + \Phi_{ext}(r) + \frac{1}{2} \Phi_{gas}(t, r) \right] dr dp_r L dL, \quad (10)$$

both of which are conserved in time. The conservation of $\mathcal{E}_{gas}$ provides a test for the accuracy of our numerical evolutions, see section V.

B. The reduced model with fixed angular momentum

A further simplification, which has been considered in Refs. [18, 21, 34], is obtained by assuming that all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum.³ In this case, the DF is of the form

$$\mathcal{F}(t, r, p_r, L) = \mathcal{F}_0(t, r, p_r) \delta(L - L_0), \quad (11)$$

and Eqs. (5,6) yield

$$\frac{\partial \mathcal{F}_0}{\partial t} + p_r \frac{\partial \mathcal{F}_0}{\partial r} - \frac{\partial V_{L_0}}{\partial r} \frac{\partial \mathcal{F}_0}{\partial p_r} = 0, \quad \rho(t, r) = \frac{2\pi L_0}{r^2} \int_{-\infty}^{\infty} \mathcal{F}_0(t, r, p_r) dp_r, \quad (12)$$

where the potential Φ_{gas} is still given by Eq. (7). Likewise, Eqs. (9,10) hold with $\mathcal{F}$ replaced by $\mathcal{F}_0$, L by L_0 , and the integral over L by the factor L_0 . Equations (12,7) constitute a Vlasov-Poisson system on a two-dimensional phase space with coordinates (r, p_r) . It describes a gas of concentric spherical shells which move in the effective potential V_{L_0} . The centrifugal term in this potential prevents the shells from reaching the center, and the gas configurations considered in this article are confined to a region of the form $0 < r_{in} \leq r \leq r_{out}$. In spite of its simplicity, the reduced model retains the two ingredients which are relevant for this article, namely the dependency of the frequency of the radial motion on the orbit, which is responsible for the mixing, and the self-gravity of the gas.

In the second situation considered in this article, $\mathcal{F}$ is a regular function of L whose support is contained in an interval $[L_1, L_2]$ with $0 < L_1 < L_2$. In this case, the phase space is three-dimensional, and the orbits with different values of L have different frequencies.

C. Dimensionless variables

In our numerical computations we use dimensionless variables, which are obtained by measuring masses in units of M , lengths in units of b and times in units of the dynamical time

$$t_{dyn} := \sqrt{\frac{b^3}{GM}}. \quad (13)$$

Accordingly, p_r is measured in units of $\sqrt{GM/b}$, L in units of $\sqrt{GMb}$, the potentials and the energy per unit mass in units of GM/b , and $\mathcal{F}$ in units of $M/(GMb)^{3/2}$. In terms of these variables, all the equations of this section retain their form with $G = M = b = 1$; in particular, the isochrone potential is $\Phi_{ext}(r) = -1/(1 + \sqrt{1 + r^2})$. The resulting system contains no free parameters, and the strength of the self-gravity is controlled by the ratio M_{gas}/M between the mass of the gas and the mass which generates the external potential. In the limit $M_{gas}/M \rightarrow 0$ the potential Φ_{gas} can be neglected, and one recovers the model analyzed in Ref. [18].

In the Kepler case $b = 0$ the external potential does not provide a length scale. However, in this case the system (5,6) is invariant with respect to the rescaling

$$(t, r, p_r, L, \mathcal{F}, \Phi) \mapsto (\alpha^{3/2}t, \alpha r, \alpha^{-1/2}p_r, \alpha^{1/2}L, \alpha^{-3/2}\mathcal{F}, \alpha^{-1}\Phi), \quad \alpha > 0, \quad (14)$$

which leaves M_{gas} unchanged. Therefore, b can be replaced with an arbitrary length in the definitions above, such that $\Phi_{ext}(r) = -1/r$, and in the reduced model this freedom can be used to fix the value of L_0 .

To give an idea of the physical time scale, we note that $M = 10^5 M_\odot$ and $b = 1$ pc, which are of the order of magnitude of the mass and of the core radius of a globular cluster [1], yield $t_{dyn} \approx 4.7 \times 10^4$ yr. As will be seen below, $2\pi t_{dyn}$ is a lower bound for the periods of the radial motion in the isochrone potential.

D. Steady states and action-angle variables

Suppose that the total potential Φ is independent of time. Then, the energy per unit mass

$$E := \frac{p_r^2}{2} + V_L(r) \quad (15)$$

³ Note that in Ref. [34] the symbol L refers to the square of the magnitude of the angular momentum.

is conserved along the trajectories of the gas particles, and it follows that any DF of the form

$$\mathcal{F}(r, p_r, L) = \psi(E, L), \quad (16)$$

with a nonnegative function ψ , is a time-independent solution of the Vlasov equation (5). In order for Eq. (16) to provide a solution of the Vlasov-Poisson system, the potential Φ_{gas} , which enters the definition of E through V_L , has to coincide with the one generated by the density of the DF (16) through Eqs. (6,7). This is a nonlinear problem for Φ_{gas} , whose numerical solution is described in section IV. In the following, a solution $(\mathcal{F}_{eq}, \Phi_{eq})$ of this problem, with Φ_{eq} the total potential and ρ_{eq} the corresponding density, will be referred to as a self-consistent steady state.

Next, we introduce action-angle variables for the radial motion in a time-independent potential Φ , following Refs. [1, 18]. In our applications, Φ is either the external potential (3) or the total potential Φ_{eq} of a self-consistent steady state. In both cases $\Phi'(r) = Gm_{tot}(r)/r^2$, where $m_{tot}(r)$ denotes the total mass contained in the sphere of radius r , including the one which generates the external potential. Since the function $r \mapsto rm_{tot}(r)$ increases strictly monotonically from 0 to ∞ , for each $L > 0$ the effective potential V_L has a unique critical point $r = r_{min}(L)$. It is determined by the equation $r^3\Phi'(r) = L^2$, it is a non-degenerate minimum of V_L , and it corresponds to a circular orbit with energy $E_{min}(L) := V_L(r_{min}(L))$. Since $V_L(r) \rightarrow \infty$ for $r \rightarrow 0$ and $V_L(r) \rightarrow 0$ for $r \rightarrow \infty$, the orbits with energies in the range $E_{min}(L) < E < 0$ are bound, and they oscillate between the turning points $r_1(E, L) < r_2(E, L)$, which are the two solutions of the equation $V_L(r) = E$.

As in Ref. [18] we define the area function

$$A(E, L) := \oint p_r dr = 2 \int_{r_1(E, L)}^{r_2(E, L)} \sqrt{2(E - V_L(r))} dr, \quad (17)$$

which represents the area enclosed by the orbit in the (r, p_r) -plane, and whose partial derivative with respect to E yields the period of the radial motion,

$$T(E, L) := \frac{\partial A}{\partial E}(E, L) = 2 \int_{r_1(E, L)}^{r_2(E, L)} \frac{dr}{\sqrt{2(E - V_L(r))}}. \quad (18)$$

The action and angle variables associated with the radial motion are

$$J := \frac{A(E, L)}{2\pi}, \quad Q := \frac{2\pi}{T(E, L)} \int_{\gamma_r} \frac{dr}{p_r}, \quad (19)$$

where γ_r denotes the segment of the orbit in the (r, p_r) -plane which connects, in the clockwise direction, the inner turning point $(r_1(E, L), 0)$ with the given point (r, p_r) . Hence, Q/Ω with $\Omega := 2\pi/T(E, L)$ is the time which has elapsed since the last passage of the particle through its pericenter; in particular, $Q = 0$ at the pericenter, $Q = \pi$ at the apocenter, and $0 < Q < \pi$ when $p_r > 0$. For each fixed value of L , the map $(r, p_r) \mapsto (Q, J)$ is symplectic, that is $dr \wedge dp_r = dQ \wedge dJ$, and it provides smooth coordinates on the set of bound orbits which are not circular. Since $T > 0$, the first relation in Eq. (19) can be inverted for each L , which yields the energy as a function $E(J, L)$ of the action variable and the angular momentum, and the frequency of the radial motion is

$$\Omega(J, L) = \frac{\partial E}{\partial J}(J, L) = \frac{2\pi}{T(E, L)}. \quad (20)$$

[(Erik) We denote the orbital frequency by Ω (it is called ω in Ref. [18]) and reserve ω for the frequency of the collective oscillation.] In terms of the variables (Q, J, L) , the volume element (8) is $8\pi^2 dQ dJ L dL$, and the Vlasov equation (5) for a time-independent potential assumes the simple form

$$\frac{\partial \mathcal{F}}{\partial t} + \Omega(J, L) \frac{\partial \mathcal{F}}{\partial Q} = 0. \quad (21)$$

In particular, the steady states (16) are precisely those DFs which are independent of the angle variable, and they can be regarded as functions of (J, L) .

For the external potential (3) the area function can be computed explicitly, see Refs. [1, 18]. In terms of the dimensionless variables introduced in the previous subsection, one finds

$$A(E, L) = 2\pi \left[\frac{1}{\sqrt{-2E}} - c(L) \right], \quad E_{min}(L) = -\frac{1}{2c(L)^2}, \quad (22)$$

with $c(L) := (L + \sqrt{L^2 + 4})/2$ for the isochrone potential and $c(L) := L$ for the Kepler potential. This yields

$$E(J, L) = -\frac{1}{2[J + c(L)]^2}, \quad \Omega(J, L) = \frac{1}{[J + c(L)]^3} = (-2E)^{3/2}. \quad (23)$$

Therefore, the frequency depends only on the energy, it decreases monotonically when J or L increase, and for each fixed L its maximum is attained in the limit $J \rightarrow 0$ of circular orbits. For the isochrone potential, $c(L) > 1$ implies that $\Omega < 1$, such that the periods of the radial motion are larger than 2π . An explicit expression for the angle variable Q in terms of (r, p_r, L) can be found in section 3.5.2 of Ref. [1]; its inversion requires the solution of an equation of Kepler's type. In contrast to this, for a self-consistent steady state the potential Φ_{eq} is only known numerically, and in this case we compute the action-angle variables by numerical quadrature of Eqs. (17,18,19), see Appendix A.

E. Observables

As in Ref. [18], we describe the macroscopic state of the gas through observables of the form

$$N_\varphi(t) := \int f(t, x, p) \varphi(x, p) d^3x d^3p, \quad (24)$$

where φ is a given test function on phase space. In this article we consider two classes of such observables.

The first one refers to the action-angle variables (Q, J) belonging to a time-independent reference potential. Unless otherwise stated, this is the total potential Φ_{eq} of the self-consistent steady state whose perturbations are analyzed, which reduces to Φ_{ext} when the self-gravity of the gas is neglected. For a spherically symmetric configuration and a test function of the form $\varphi = a(Q)B(J, L)$, with real-valued functions a and B , Eqs. (8,24) and the expansion of a in a Fourier series yield

$$N_\varphi(t) = \sum_{n \in \mathbb{Z}} \hat{a}_n^* h_n(t), \quad \hat{a}_n := \frac{1}{2\pi} \int_0^{2\pi} a(Q) e^{-inQ} dQ, \quad (25)$$

with the functions

$$h_n(t) := 8\pi^2 \int_0^\infty \int_0^\infty \int_0^{2\pi} \mathcal{F}(t, Q, J, L) B(J, L) e^{-inQ} dQ dJ L dL, \quad n \in \mathbb{Z}. \quad (26)$$

In the reduced model, B depends only on J and Eq. (26) holds with $\mathcal{F}$ replaced by $\mathcal{F}_0$ and the integral over L by the factor L_0 . Since $\mathcal{F}$ is real, $h_{-n} = h_n^*$, such that it is sufficient to consider $n \geq 0$. For a steady state h_n vanishes for all $n \neq 0$, whereas h_0 is equal to the mass of the gas weighted with the function B . Therefore, the functions $h_n(t)$ with $n \neq 0$ measure the departure of the gas configuration from a steady state, and their decay in time is the signature of the relaxation process we are interested in. We stress that this property relies on the use of the action-angle variables which belong to the total potential Φ_{eq} . If those of the external potential are used instead, a self-consistent steady state gives rise to nonvanishing, time-independent values of h_n with $n \neq 0$, as we show in section V.

The second class of observables consists of the perturbation of the gravitational potential,

$$\delta\Phi(t, r) := \Phi(t, r) - \Phi_{eq}(r) = -4\pi G \int_0^\infty \frac{\rho(t, s) - \rho_{eq}(s)}{\max\{r, s\}} s^2 ds, \quad (27)$$

where we have used Eq. (7). For each fixed r this is the difference between the values of the observable (24) with test function $\varphi(x, p) = -G/\max\{r, |x|\}$ which belong to the DF of the gas and to the one of the steady state. In contrast to the functions $h_n(t)$, the quantity $\delta\Phi$ does not refer to any particular coordinates on phase space. The results in Refs. [34, 35] refer to the decay of the force field $-\partial\delta\Phi/\partial r$ which is generated by it, or of the time derivative of this field. To measure its magnitude we use the root mean square

$$\|\delta\Phi(t)\| := \left(\frac{1}{r_b - r_a} \int_{r_a}^{r_b} \delta\Phi(t, r)^2 dr \right)^{1/2} \quad (28)$$

over an interval $[r_a, r_b]$ which covers the region occupied by the gas.

III. MIXING AND LINEAR RESPONSE

In this section we first review the mixing phenomenon in the absence of self-gravity and discuss the rate at which the observables introduced in the previous section decay. Next, we linearize the Vlasov-Poisson system about a self-consistent steady state and formulate, in terms of the action-angle variables of the latter, the criterion for the existence of oscillating modes. This leads to the gain $\lambda(\omega)$ of the self-consistent loop mentioned in the introduction, whose behavior at the edge of the frequency band is discussed at the end of the section. The derivations in this section are formal; for rigorous statements we refer the reader to Refs. [18, 20] regarding the mixing property and to Refs. [29, 30, 33, 34] regarding the linearized system.

A. Mixing in the absence of self-gravity

When the self-gravity of the gas is neglected, $\Phi = \Phi_{ext}$ is independent of time, and the solution of Eq. (21) is

$$\mathcal{F}(t, Q, J, L) = F(Q - \Omega(J, L)t, J, L), \quad (29)$$

where the function F , which is 2π -periodic in its first argument, describes the initial DF. Expanding F in a Fourier series with respect to the angle variable,

$$F(Q, J, L) = \sum_{n \in \mathbb{Z}} \hat{F}_n(J, L) e^{inQ}, \quad \hat{F}_n(J, L) := \frac{1}{2\pi} \int_0^{2\pi} F(Q, J, L) e^{-inQ} dQ, \quad (30)$$

and introducing Eq. (29) into Eq. (26) one obtains

$$h_n(t) = 16\pi^3 \int_0^\infty \int_0^\infty \hat{F}_n(J, L) B(J, L) e^{-in\Omega(J, L)t} dJ L dL. \quad (31)$$

Hence, h_0 is constant in time, whereas for $n \neq 0$ the function $h_n(t)$ is a superposition of oscillations whose frequencies $n\Omega(J, L)$ vary from one orbit to the other. Provided that the gradient of Ω with respect to (J, L) is different from zero for almost all (J, L) (in the reduced model, provided that $\partial\Omega/\partial J \neq 0$ for almost all J), it follows from the Riemann-Lebesgue lemma and its generalizations (cf. Refs. [18, 20]) that $h_n(t) \rightarrow 0$ for $t \rightarrow \infty$ and all $n \neq 0$. As a consequence, the observables (25) converge to the ones which belong to the angle-averaged DF $\hat{F}_0(J, L)$, which is a steady state according to the remark below Eq. (21). This is the phase space mixing property. For the isochrone and the Kepler potentials, Eq. (23) gives $\partial\Omega/\partial J = -3[J + c(L)]^{-4} < 0$, such that the non-degeneracy condition is satisfied.

The rate at which $h_n(t)$ decays is not universal. It depends on the regularity of the function $\hat{F}_n B$ and on its behavior at the boundary of its support. If this function is smooth and vanishes together with all its derivatives at the boundary, repeated integration by parts in Eq. (31) shows that $h_n(t)$ decays faster than any inverse power of t . A different situation, which is the relevant one for the steady states considered in this article, arises in the reduced model when the DF has its support in an interval $0 \leq J \leq J_{max}$ and vanishes like a power at the outer edge, such that

$$\hat{F}_n(J) B(J) = (J_{max} - J)^k g(J) \quad (32)$$

with an exponent $k > 0$ and a smooth function g satisfying $g(J_{max}) \neq 0$. Suppose that $\Omega'(J) < 0$ for all $0 < J \leq J_{max}$, as is the case in all the examples of this article, and that g vanishes sufficiently fast for $J \rightarrow 0$. Then, the behavior of the integral in Eq. (31) for large times is determined by the endpoint $J = J_{max}$ (see, for instance, chapter 6 in Ref. [52]), and for $n \geq 1$ one finds

$$h_n(t) \simeq 16\pi^3 L_0 \Gamma(k+1) g(J_{max}) \frac{e^{-in\Omega_{min}t}}{[in|\Omega'(J_{max})|t]^{k+1}}, \quad t \rightarrow \infty, \quad (33)$$

where $\Omega_{min} := \Omega(J_{max})$ is the frequency of the outermost orbit and Γ denotes the Gamma function. Therefore, an edge at which the DF vanishes with the exponent k gives rise to an algebraic tail which is proportional to $t^{-(k+1)}$ and which oscillates with the frequency $n\Omega_{min}$: the less regular the DF is at the edge, the slower is the decay. Algebraic decay laws in related settings have been obtained in Refs. [26, 49].

When the angular momenta of the gas particles are distributed, the integration over L in Eq. (31) provides an additional source of decay, since Ω depends on L as well. For the isochrone and the Kepler potentials, Eq. (23) yields $\partial\Omega/\partial L = c'(L) \partial\Omega/\partial J$ with $1/2 < c'(L) \leq 1$, such that a dispersion in L is almost as efficient for the mixing as a dispersion of the same magnitude in J . This effect is analyzed in section VII.

B. Linearization about a self-consistent steady state

Let $(\mathcal{F}_{eq}, \Phi_{eq})$ be a self-consistent steady state, and let (Q, J) be the action-angle variables which belong to the potential Φ_{eq} , such that $\mathcal{F}_{eq}$ is a function of (J, L) only. We write

$$\mathcal{F} = \mathcal{F}_{eq} + \delta\mathcal{F}, \quad \Phi = \Phi_{eq} + \delta\Phi, \quad (34)$$

with $\delta\Phi$ the perturbation of the potential defined in Eq. (27). The motion of the gas particles is determined by the Hamiltonian $H = E(J, L) + \delta\Phi(t, r(Q, J, L))$, where $r(Q, J, L)$ denotes the radius as a function of the action-angle variables. Since for each fixed L the variables (Q, J) are canonical, the Vlasov equation (5) is equivalent to

$$\frac{\partial\delta\mathcal{F}}{\partial t} + \Omega \frac{\partial\delta\mathcal{F}}{\partial Q} - \frac{\partial\mathcal{F}_{eq}}{\partial J} \frac{\partial\delta\Phi}{\partial Q} = \frac{\partial\delta\Phi}{\partial Q} \frac{\partial\delta\mathcal{F}}{\partial J} - \frac{\partial\delta\Phi}{\partial J} \frac{\partial\delta\mathcal{F}}{\partial Q}, \quad (35)$$

where no approximation has been made so far. The right-hand side is quadratic in the perturbation, and neglecting it yields the linearized equation

$$\frac{\partial\delta\mathcal{F}}{\partial t} + \Omega(J, L) \frac{\partial\delta\mathcal{F}}{\partial Q} - \frac{\partial\mathcal{F}_{eq}}{\partial J}(J, L) \frac{\partial\delta\Phi}{\partial Q} = 0. \quad (36)$$

The potential is determined by Eqs. (6,7,8), which give

$$\delta\Phi(t, r) = -8\pi^2 G \int_0^\infty \int_0^\infty \int_0^{2\pi} \frac{\delta\mathcal{F}(t, Q', J', L')}{\max\{r, r(Q', J', L')\}} dQ' dJ' dL'. \quad (37)$$

The second term on the left-hand side of Eq. (36) describes the transport along the orbits of the steady state discussed in the previous subsection. The third term describes the reaction of the steady state to the perturbation of the potential: the latter changes the action variables of the gas particles at the rate $\dot{J} = -\partial\delta\Phi/\partial Q$, and this displaces mass along the gradient of $\mathcal{F}_{eq}$. Through this term, which is absent when self-gravity is neglected, all the orbits are coupled to each other.

Since the support of $\mathcal{F}_{eq}$ is bounded, the frequencies $\Omega(J, L)$ of the orbits which are populated by the steady state fill an interval $[\Omega_{min}, \Omega_{max}]$ with $0 < \Omega_{min} < \Omega_{max}$, which will be referred to as the frequency band in what follows. Without the third term in Eq. (36), the n -th Fourier coefficient of $\delta\mathcal{F}$ with respect to Q oscillates with the frequencies $n\Omega(J, L)$, which lie in the interval $n[\Omega_{min}, \Omega_{max}]$. These intervals constitute the essential spectrum of the linearized system, which is not altered by the self-gravity of the perturbation [29, 34].

C. Oscillating modes and the gain of the self-consistent loop

We look for solutions of Eqs. (36,37) of the form

$$\delta\mathcal{F}(t, Q, J, L) = a(Q, J, L)e^{-i\omega t} + c.c., \quad \delta\Phi(t, r) = \phi(r)e^{-i\omega t} + c.c., \quad (38)$$

with a real frequency $\omega > 0$, where *c.c.* denotes the complex conjugate of the preceding term. Such a solution describes a perturbation which oscillates without being damped, and it will be called an oscillating mode in what follows. We expand

$$a(Q, J, L) = \sum_{n \in \mathbb{Z}} a_n(J, L)e^{inQ}, \quad \phi(r(Q, J, L)) = \sum_{n \in \mathbb{Z}} \phi_n(J, L)e^{inQ}. \quad (39)$$

Since the radius is the same at the two points of an orbit with angle variables Q and $2\pi - Q$, the coefficients of the potential satisfy $\phi_{-n} = \phi_n$ and are given by

$$\phi_n(J, L) = \frac{1}{2\pi} \int_0^{2\pi} \phi(r(Q, J, L)) \cos(nQ) dQ. \quad (40)$$

Equation (36) yields

$$[n\Omega(J, L) - \omega] a_n(J, L) = n \frac{\partial\mathcal{F}_{eq}}{\partial J}(J, L) \phi_n(J, L), \quad n \in \mathbb{Z}, \quad (41)$$

which implies that $a_0 = 0$. From now on we assume that ω lies below the frequency band, that is $0 < \omega < \Omega_{min}$.⁴ In this case none of the gas particles is in resonance with the perturbation, in the sense that $n\Omega(J, L) \neq \omega$ for all $n \in \mathbb{Z}$ and all the orbits populated by the steady state, and Eq. (41) can be solved for a_n . By the same symmetry argument as above, the potential generated by the perturbation depends only on the combinations $b_n := a_n + a_{-n}$ with $n \geq 1$, for which Eq. (41) gives

$$b_n(J, L) = R_n(J, L; \omega) \phi_n(J, L), \quad R_n(J, L; \omega) := \frac{2n^2 \Omega(J, L)}{n^2 \Omega(J, L)^2 - \omega^2} \frac{\partial \mathcal{F}_{eq}}{\partial J}(J, L). \quad (42)$$

On the other hand, introducing the ansatz (38) into Eq. (37) and using Eq. (40) one obtains

$$\phi_n(J, L) = \sum_{n'=1}^{\infty} \int_0^{\infty} \int_0^{\infty} M_{nn'}(J, L; J', L') b_{n'}(J', L') dJ' L' dL', \quad (43)$$

with the kernel

$$M_{nn'}(J, L; J', L') := -4\pi G \int_0^{2\pi} \int_0^{2\pi} \frac{\cos(nQ) \cos(n'Q')}{\max\{r(Q, J, L), r(Q', J', L')\}} dQ dQ'. \quad (44)$$

Equations (42,43) can be written in the compact form

$$\phi = \mathcal{K}(\omega) \phi, \quad \mathcal{K}(\omega) := M R(\omega), \quad (45)$$

where $\phi = (\phi_n)_{n \geq 1}$, $R(\omega)$ denotes the multiplication with the functions $R_n(\cdot; \omega)$, and M is the integral operator defined by the right-hand side of Eq. (43). The operator $\mathcal{K}(\omega)$ maps a perturbation of the potential with frequency ω to the potential which is generated by the response of the gas to this perturbation. Therefore, an oscillating mode with frequency ω exists if and only if one is an eigenvalue of $\mathcal{K}(\omega)$. In the reduced model, Eqs. (40–44) hold with $L = L' = L_0$, the function $\mathcal{F}_{eq}$ replaced by $\mathcal{F}_{0,eq}(J)$, and the integral over L' by the factor L_0 .

The operators M and $R(\omega)$ have the following properties. First, with respect to the inner product

$$\langle b, c \rangle := \sum_{n=1}^{\infty} \int_0^{\infty} \int_0^{\infty} b_n(J, L) c_n(J, L) dJ L dL, \quad (46)$$

the operator M is symmetric and negative, since for real-valued b one finds

$$-\langle b, Mb \rangle = \frac{1}{64\pi^4 G} \int_{\mathbb{R}^3} |\nabla \delta \Phi_b|^2 d^3x \geq 0, \quad (47)$$

where $\delta \Phi_b$ denotes the potential which is obtained from Eq. (37) when $\delta \mathcal{F}$ is replaced with $\sum_{n \geq 1} b_n(J, L) \cos(nQ)$. Second, suppose that $\partial \mathcal{F}_{eq} / \partial J < 0$ in the interior of the support of the steady state. Since $\partial \mathcal{F}_{eq} / \partial J = \Omega \partial \psi / \partial E$, this means that the function ψ in Eq. (16) decreases with the energy, which is Antonov's stability condition [1, 34], and it holds for all the steady states considered in this article. Then, $R_n(J, L; \omega) < 0$ for all $0 \leq \omega < \Omega_{min}$, and $\mathcal{K}(\omega) = (-M)|R(\omega)|$ has the same nonvanishing eigenvalues as the symmetric and nonnegative operator

$$S(\omega) := |R(\omega)|^{1/2} (-M) |R(\omega)|^{1/2}. \quad (48)$$

We denote by $\lambda(\omega)$ the largest eigenvalue of $S(\omega)$ and call it the gain of the self-consistent loop. Since $|R_n(J, L; \omega)|$ increases with ω , the function $\lambda(\omega)$ is nondecreasing on the interval $[0, \Omega_{min})$, and we define its value at the edge of the band by

$$\lambda_{edge} := \lim_{\omega \rightarrow \Omega_{min}} \lambda(\omega) \in (0, \infty]. \quad (49)$$

⁴ When $\Omega_{max} < 2\Omega_{min}$ there are further gaps between the intervals $n[\Omega_{min}, \Omega_{max}]$. We do not consider frequencies lying in these gaps, since in all our examples the collective oscillation emerges below the band.

Third, the same construction applies to solutions which grow exponentially in time, $\delta\mathcal{F} \propto e^{\sigma t}$ with $\sigma > 0$, for which ω^2 is replaced with $-\sigma^2$ in Eq. (42). In this case $|R_n|$ decreases to zero when σ increases from zero to infinity, such that a growing solution exists if and only if $\lambda(0) > 1$. Hence, the linear stability of the steady state with respect to spherically symmetric perturbations requires that $\lambda(0) \leq 1$. For all the steady states of the reduced model which are analyzed in section VI we find $\lambda(0) < 0.6$.

Combining these properties, we arrive at the following criterion for a linearly stable steady state with $\lambda(0) < 1$. If $\lambda_{edge} > 1$, there exists an oscillating mode whose frequency ω_d lies below the frequency band and is determined by the equation

$$\lambda(\omega_d) = 1, \quad 0 < \omega_d < \Omega_{min}. \quad (50)$$

If $\lambda_{edge} < 1$, there are no oscillating modes with frequencies below the band. Further modes would appear if other eigenvalues of $S(\omega)$ exceeded one. This does not occur in our examples, in which the second largest eigenvalue remains smaller than 0.26 up to the edge of the band. Note that $1/\lambda_{edge}$ is the factor by which the perturbation of the potential in Eq. (36) would have to be multiplied for an oscillating mode to appear at the edge of the band; in this sense, λ_{edge} measures the strength of the self-gravity of the perturbation relative to the one that is required to prevent its mixing.

This criterion is a formulation in terms of action-angle variables of the Birman-Schwinger principle of Refs. [29, 30, 32, 34]. In fact, $\mu > 0$ is an eigenvalue of $\mathcal{K}(\omega)$ if and only if the linearized system in which $\delta\Phi$ is multiplied by the factor $1/\mu$ possesses an oscillating mode with frequency ω , and this is the property which characterizes the eigenvalues of the Birman-Schwinger operator of Ref. [34] with spectral parameter ω^2 . Therefore, $\lambda(\omega)$ coincides with the norm of the operator of Mathur defined in that reference, and the interval $0 < \omega < \Omega_{min}$ corresponds to the principal gap of the spectrum. In Refs. [29, 30, 34] the criterion is used to prove the existence or the absence of oscillating modes for certain classes of steady states. In section IV we describe a numerical method for the computation of $\lambda(\omega)$, which allows us to evaluate the criterion for steady states of arbitrary mass. [(Erik) As far as we could determine, this operator has not been evaluated numerically in the literature; Kunze's lecture notes [31] and Straub's thesis list it as an open task. Mathur's article [32] and Kunze's book [30] have to be checked before claiming this.]

D. Behavior at the edge of the frequency band

The value of λ_{edge} depends strongly on the way in which the DF of the steady state vanishes at the boundary of its support. To explain this, consider the reduced model and a steady state whose DF has a cutoff at the energy E_0 and behaves like

$$\mathcal{F}_{0,eq} \simeq \alpha(E_0 - E)^k, \quad E \rightarrow E_0, \quad (51)$$

with constants $\alpha > 0$ and $k > 1/2$, as is the case for the polytropes considered in Ref. [34]. Let J_{max} be the action variable of the orbit with energy E_0 , such that $\Omega_{min} = \Omega(J_{max})$ when Ω is a decreasing function of J . Then $\partial\mathcal{F}_{0,eq}/\partial J$ is proportional to $(J_{max} - J)^{k-1}$ near the edge, and Eq. (42) with $n = 1$ and $\omega = \Omega_{min}$ yields

$$|R_1(J; \Omega_{min})| \simeq \frac{1}{\Omega(J) - \Omega_{min}} \left| \frac{\partial\mathcal{F}_{0,eq}}{\partial J}(J) \right| \propto (J_{max} - J)^{k-2}, \quad J \rightarrow J_{max}. \quad (52)$$

This function is integrable at $J = J_{max}$ if and only if $k > 1$, which leads to the following two cases.

If $k > 1$, λ_{edge} is finite. Since $R(\omega)$ is proportional to the DF of the steady state, λ_{edge} is proportional to M_{gas}/M as long as this ratio is small enough for the contribution of Φ_{gas} to Φ_{eq} to be negligible. Therefore, for a gas of small mass one has $\lambda_{edge} < 1$ and there is no oscillating mode, in accordance with the damping result in Ref. [34]. When the mass of the gas increases, λ_{edge} grows, and an oscillating mode appears when it reaches the value one. The corresponding threshold mass is computed in section VI. Close to the edge one finds $\lambda_{edge} - \lambda(\omega) \propto (\Omega_{min} - \omega)^{\min\{k-1, 1\}}$, with a logarithmic correction when $k = 2$. Hence, slightly above the threshold the frequency of the mode satisfies $\Omega_{min} - \omega_d \propto (\lambda_{edge} - 1)^{1/(k-1)}$ when $1 < k < 2$, such that the closer k lies to one, the closer the frequency of the mode stays to the edge of the band.

If $1/2 < k \leq 1$, $\lambda(\omega)$ diverges for $\omega \rightarrow \Omega_{min}$, like $(\Omega_{min} - \omega)^{k-1}$ when $k < 1$ and like $|\log(\Omega_{min} - \omega)|$ when $k = 1$. Hence, $\lambda_{edge} = \infty$ and an oscillating mode exists for any mass of the gas, as proven in Ref. [34] for small masses. However, since the factor which multiplies the divergent term is proportional to M_{gas}/M , Eq. (50) implies that, to leading order in the limit $M_{gas}/M \rightarrow 0$,

$$\Omega_{min} - \omega_d \propto \left(\frac{M_{gas}}{M} \right)^{\frac{1}{1-k}} \quad (k < 1), \quad \Omega_{min} - \omega_d \propto \exp\left(-C \frac{M}{M_{gas}}\right) \quad (k = 1), \quad (53)$$

with a positive constant C . Therefore, when the mass of the gas is small the frequency of the mode lies extremely close to the edge of the band. Since a time of the order of $(\Omega_{min} - \omega_d)^{-1}$ is required to distinguish the mode from the response of the orbits close to the edge, which oscillates with a frequency close to Ω_{min} and decays algebraically (cf. Eq. (33)), in this regime the mode cannot be observed in a time evolution of practical duration.

When the angular momenta of the gas particles are distributed, the support of the steady state in the (J, L) -plane is two-dimensional, and in general the minimum Ω_{min} of the frequency is attained only at isolated points of its boundary. The additional integration over L in Eq. (43) then weakens the singularity of R_1 at the edge of the band, such that λ_{edge} can be finite also for $k \leq 1$, in which case the oscillating modes of the reduced model disappear when the mass of the gas is small enough. This is analyzed in section VII. [(Erik) The precise statement depends on the family of steady states: for a cutoff in the energy and the isochrone or Kepler potential, Ω depends on L at fixed energy only through the self-gravity. To be settled together with section VII.]

Finally, we comment on the case $\lambda_{edge} < 1$, in which the linearized system has no oscillating mode below the band. As we show in section VI, in this case the perturbation decays. When the self-gravity of the gas is not too weak, the response to an initial perturbation is dominated at intermediate times by a damped oscillation proportional to $e^{-i\omega t - \gamma t}$ with a frequency ω inside the band and a damping rate $\gamma > 0$, which is followed at late times by an algebraic tail of the form (33). The damped oscillation is the analogue of Landau damping in a plasma [27]. It does not correspond to an eigenvalue of the linearized system; instead, $\omega - i\gamma$ is a zero of the analytic continuation of the function $\omega \mapsto \det[1 - \mathcal{K}(\omega)]$ from the upper to the lower half of the complex plane, see for instance Refs. [1, 39, 40]. We do not compute this continuation in this article. Instead, we determine ω and γ by fitting the time series which are obtained from the numerical solution of the linearized equations and from the PIC code, as described in section IV.

IV. NUMERICAL METHODS

[(Erik) To be written: particle-in-cell method, equilibria and initial data, linearized solver, evaluation of the gain.]

V. CODE VALIDATION

[(Erik) To be written.]

VI. RESULTS FOR FIXED ANGULAR MOMENTUM

[(Erik) To be written.]

VII. RESULTS FOR A DISTRIBUTION OF ANGULAR MOMENTA

[(Erik) To be written.]

VIII. CONCLUSIONS

[(Erik) To be written.]

Acknowledgments

[(Erik) To be written.]

Appendix A: Numerical computation of the action-angle variables

[(Erik) To be written.]

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